Capability Statement: Open Vision Technology, LLC.

CAGE: 18NH2 | **UEI:** U6GJVDKFSYG1 | **SBA Certifications:** EDWOSB, WOSB **| Point of Contact:** Denise Lin | **Email:** denise@openvisiontech.com **Website:** openvisiontech.com | **Address:** 8 Thompson Pond Road, Stoneham, MA 02180

Company Overview

Open Vision Technology, LLC provides high-performance full-stack software solutions and tactical AI middleware for autonomous robotics and Edge AI applications. Our flagship software product, **Uli SDK**, serves as the connective software layer bridging physical telemetry, edge platforms, and Agentic AI reasoning engines. By dynamically transforming physical assets—including mobile robotic fleets, tactical sensors, and legacy payloads—into a structured **Knowledge Graph Context Layer**, Uli SDK accelerates autonomous command, semantic state estimation, and interoperability across multi-domain ecosystems.

Core Competencies

- **Multi-Domain Heterogeneous Asset Integration:** Seamlessly bridging disparate robotic platforms, autonomous systems, legacy hardware, and tactical edge sensors across multi-domain environments into a unified software architecture.
- **Dynamic Context & Knowledge Graph Synthesis:** Automated discovery and semantic modeling of asset identity, functional capabilities, and live telemetry feeds into a structured Knowledge Graph, constructing dynamic world models for AI agent reasoning and command execution.
- **Human-Machine Teaming via Cursor-on-Target (CoT):** Real-time, bi-directional situational awareness sharing through native CoT protocol support, establishing a unified Common Operational Picture (COP) for both human operators (e.g., TAK ecosystem) and AI agents.
- **Autonomous Command & Execution Pipelines:** Translating higher-level AI agent reasoning into high-fidelity control parameters, task allocations, and mission execution protocols with zero human-in-the-loop latency bottlenecks.
- **Agent-to-UI (A2UI) Dynamic Visualization:** Utilizing Dart-FFI and Flutter integration to dynamically generate operator dashboards and operational controls based on real-time Knowledge Graph asset discovery.
- **Edge-Optimized Middleware Architecture:** High-throughput, low-latency bare-metal C++ framework designed to execute directly on edge compute nodes (e.g., NVIDIA Jetson) without relying on persistent cloud connectivity.

Differentiators

- **NGC2 & JADC2 Interoperability:** Engineered to align with the U.S. Army's Next Generation Command and Control (NGC2) vision and joint all-domain frameworks, providing a common tactical context fabric for heterogeneous manned and unmanned systems.
- **Automated Asset & Capability Discovery:** Uli SDK's discovery layer eliminates manual middleware mapping, reducing integration latency by **40%+** when onboarding new sensors or robotic payloads.
- **Hardware & Sensor Agnostic:** Scalable architecture deployed on edge hardware ranging from our internal 3-wheel **Uli Kaya** mobile reference platform to fixed tactical telemetry sensors and vehicle buses.
- **Real-Time Edge-to-Agent Pipeline:** Built in optimized C++ and Dart-FFI to deliver low-latency telemetry ingestion and execution directly on tactical edge computers without relying on cloud availability.

Past Performance

Uli SDK Implementation:

- Developed and deployed a high-performance discovery-driven middleware that bridges the gap between traditional robotic infrastructures and Agentic AI ecosystems.
- Enabled connected assets to dynamically export identity, functional capabilities, and telemetry into a structured **Knowledge Graph-driven Context Layer**.

Subcontractor for Defense Primes:

- Served as a specialized subcontractor for defense primes, contributing to the development of sophisticated command and control (C2) applications for multiple Department of Defense (DoD) **Programs of Record (PoR)**.
- Implemented and integrated industry-standard communication protocols and architectural frameworks, including:
  - **SAE AS4 (JAUS)**: Unmanned systems interoperability and message profiling.
  - **J1939**: Vehicle bus instrumentation and diagnostics for heavy-duty ground assets.
  - **DDS (Data Distribution Service)**: High-performance, real-time data bus architectures for distributed systems.
  - **ROS**: Modular software integration for robotic sensing and perception.

Project Uli Kaya (Autonomous Reference Platform):

- Designed, built, and deployed the Uli SDK onto Uli Kaya, a proprietary 3-wheeled omnidirectional autonomous robot platform running Jetson-based compute.
- Successfully demonstrated end-to-end edge AI command execution, real-time telemetry extraction, and dynamic A2UI dashboard rendering.

Corporate Data

- **Socio-Economic Status:** Economically Disadvantaged Woman-Owned Small Business (EDWOSB) & Woman-Owned Small Business (WOSB) Certified
- **CAGE Code:** 18NH2 | **UEI:** U6GJVDKFSYG1
- **NAICS Codes:**
  - **541511:** Custom Computer Programming Services *(Primary)*

  - **541715:** Research and Development in the Physical, Engineering, and Life Sciences
  - **334511:** Search, Detection, Navigation, Guidance, Aeronautical, and Nautical System/Instrument Manufacturing
- **PSC Codes:**
  - **DA01:** IT and Telecom - Business Application/Application Development Support
  - **AC13:** R&D - Defense System: Electronics/Communication Equipment (Advanced Development)